Analysis of Complex Sample Survey Data

SURV/SURVMETH 701

Lecture Notes: Module 5

Replication Methods for Variance Estimation

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Alternatives to Taylor Series Linearization

- The linearization technique is useful if the estimate can be expressed as a function of sample totals; readily available in Stata, SAS, and other software
- Linearization requires analytic manipulations, computation of derivatives
- **Not directly suitable for percentiles such as the median or functions of percentiles (non-smooth functions)**
- Replication techniques are useful for almost any type of estimate
- Empirical comparisons of the different techniques: Kish and Frankel (1974), Kovar, Rao and Wu (1988), Korn and Graubard (1999) (See Canvas)

Steps in Replicated Methods

1. Create replicates using sampled cases (unique to each method). Software uses sampling error codes provided in the data set.
   – Jackknife Repeated Replication (JRR) replicate sample: leave out one PSU, and re-weight cases in "deletion stratum".
   – Balanced Repeated Replication (BRR) replicate sample: given two PSUs per stratum, leave out one PSU from each stratum (based on a "balanced" Hadamard matrix to eliminate covariances when estimating variances), and re-weight cases (see ASDA, Chapter 3).

2. Create revised weight for replicate sample.

3. Compute weighted estimates of population statistic of interest for each replicate using replicate weight.

4. Apply replicated variance estimation formula to derive standard errors.

5. Construct confidence intervals (or hypothesis tests) based on estimated statistics, standard errors, and correct degrees of freedom (generally use the previously discussed a – H approximation).

Replication-based Variance Estimation: Options in Stata

- **Jackknife Repeated Replication (JRR)**
  - Stata: Use `svy, vce(jackknife)`: commands
- **Balanced Repeated Replication (BRR)**
  - Stata: Use `svy brr, hadamard(…)`: commands
  - Requires two cluster codes per stratum!
- **Bootstrapping**
  - Stata: see Kolenikov (2010, Stata Journal).
  - Example:

```
. svyset [pw=finalwgt], vce(bootstrap) bsrweight(bw*)
. svy: logistic highbp height weight age female
```

Replicate Weighting

- JRR, BRR, and bootstrap programs adjust the values of the sample selection weights to create "replicate weights" using the above procedures. Default - no adjustment to original survey weights.
- If nonresponse adjustment and post-stratification adjustments are included in the final survey weight, survey statisticians advocate that these be recomputed for each sample replicate.
- WesVar PC (Westat, Inc.) assists in developing replicate weight adjustments that reflect nonresponse and post-stratification.
- **Replicate weights** are added to data set (one per replicate):

  w=[w(1), …,w(K)]

Simple Data Example (4 Strata, 2 PSUs/Stratum)

Stratum	PSU (Cluster)	Case	y_i	w_i
1	1	1	.58	1
	1	2	.48	2
1	2	1	.42	2
	2	2	.57	2
2	1	1	.39	1
	1	2	.46	2
2	2	1	.50	2
	2	2	.21	1
3	1	1	.39	1
	1	2	.47	2
3	2	1	.44	1
	2	2	.43	1
4	1	1	.64	1
	1	2	.55	1
4	2	1	.47	2
	2	2	.50	2

JRR: Constructing Replicates, Replicate Weights

- There are several approaches to constructing the JRR replicates. The "Delete One" method that is often the default in survey analysis software is illustrated here.

- For the Delete One JRR variance estimator, each stratum will contribute a_h replicates, where for each replicate, one of the clusters in a single stratum is deleted.

- In our data example, H=4 and a_h= 2 for h = 1,…,4, so the total number of required JRR replicates for the Delete One JRR is:

$$\sum_{h=1}^{4} a_h = 8$$

JRR: Constructing Replicates, Replicate Weights

- Suppose there are H strata with a_h clusters in the h^{th} stratum. H=4, a_h=2 in example data set.

- A JRR replicate is constructed by deleting one PSU from one stratum. The first replicate leaves out the 1st PSU in Stratum 1.

- The replicate weight for this first replicate multiplies the weights for remaining cases in the "deletion stratum" by a factor of $a_h/[a_h - 1]$. This equals 2 in our example. Weight values for Replicate 1 remain unchanged for cases in all other strata.

JRR Replicate 1: Data Example

Stratum	PSU (Cluster)	Case	y_i	$w_{i,rep}$
1	1	1	.	.
	1	2	.	.
1	2	1	.42	2x2
	2	2	.57	2x2
2	1	1	.39	1
	1	2	.46	2
2	2	1	.50	2
	2	2	.21	1
3	1	1	.39	1
	1	2	.47	2
3	2	1	.44	1
	2	2	.43	1
4	1	1	.64	1
	1	2	.55	1
4	2	1	.47	2
	2	2	.50	2

JRR Replicate 2 (of 8): Data Example

Stratum	PSU (Cluster)	Case	y_i	$w_{i,rep}$
1	1	1	.58	1x2
	1	2	.48	2x2
1	2	1	.	.
	2	2	.	.
2	1	1	.39	1
	1	2	.46	2
2	2	1	.50	2
	2	2	.21	1
3	1	1	.39	1
	1	2	.47	2
3	2	1	.44	1
	2	2	.43	1
4	1	1	.64	1
	1	2	.55	1
4	2	1	.47	2
	2	2	.50	2

JRR: Constructing Estimates

Here, eight replicates are created and the weighted estimate:

$\hat{q}_k = \bar{y}_k^w = \dfrac{\sum_{i \in rep}^{n_{rep}} y_i w_{i,rep}}{\sum_{i \in rep}^{n_{rep}} w_{i,rep}}$ is computed for each of k=1,...,8 replicates

$\hat{q}_{1(\text{del } 1\_1)} = 0.47240$; $\hat{q}_{3(\text{del } 2\_1)} = 0.46958$; $\hat{q}_{5(\text{del } 3\_1)} = 0.47434$; $\hat{q}_{7(\text{del } 4\_1)} = 0.46615$

$\hat{q}_{2(\text{del } 1\_2)} = 0.47521$; $\hat{q}_{4(\text{del } 2\_2)} = 0.47791$; $\hat{q}_{6(\text{del } 3\_2)} = 0.47320$; $\hat{q}_{8(\text{del } 4\_2)} = 0.48272$;

$\hat{q} = \dfrac{\sum_{i=1}^{n} y_i w_i}{\sum_{i=1}^{n} w_i}$ the full sample estimate is also computed.

$\hat{q}=0.47375$

JRR Variance Estimation: Delete 1

- If stratum and cluster codes are **available** in the data set:

$$\text{var}_{JRR}(\hat{q}) = \sum_{h=1}^{H} \frac{a_h - 1}{a_h} \sum_{\alpha=1}^{a_h} (\hat{q}_{(h\alpha)} - \hat{q})^2$$

where

$\hat{q}_{(h\alpha)} =$ weighted estimate of Q for replicate where cluster α in stratum h has been deleted;

$df = a - H$

- If **only (*a*) replicate weights** are available in the data set:

$$\text{var}_{JRR}(\hat{q}) = \frac{a-1}{a} \sum_{k=1}^{a} (\hat{q}_k - \hat{q})^2$$

$df = a - 1$

JRR: Estimating the Sampling Variance

Since stratum and cluster codes are **available** in the data example

$$var_{JRR_{DelOne}}(\hat{q}) = \sum_{h=1}^{H} \frac{a_h - 1}{a_h} \sum_{\alpha=1}^{a_h} \left(\hat{q}_{(h\alpha)} - \hat{q}\right)^2$$

$$= \sum_{h=1}^{4} \frac{1}{2} \sum_{\alpha=1}^{2} \left(\hat{q}_{(h\alpha)} - \hat{q}\right)^2$$

$$= .5 \sum_{k=1}^{8} (\hat{q}_k - \hat{q})^2$$

JRR: Estimating the Sampling Variance

$$var_{JRR_{DelOne}}(\hat{q}) = 0.5\sum_{k=1}^{8}(\hat{q}_k - \hat{q})^2 = .0000801449$$

$$se_{JRR}(\bar{y}^w) = \sqrt{.0000801449} = .008952$$

$$CI(\bar{y}^w) = .47375 \mp t_{1-\frac{\alpha}{2},4} \cdot .008952$$
$$= .47375 \mp 2.7764 \cdot .008952$$
$$= (.4489, .4986)$$

Alternative for Replication Variance Estimates

- <u>Stata Code</u>

```
svyset [pweight=wgtvar], vce(jackknife) jkrweight(wt*)
svy: mean varname
```

Half Sample Replicates

- Assume a paired selection design (2 PSUs per stratum design)
- A half sample is defined by choosing one PSU from each stratum.
- A complement of a half sample is made up of all those PSUs not in the half sample. A complement is also a half sample.
- There are 2^H possible half samples and their complements. We only need H half samples for variance estimation. (Why?)

Half-Sample Replication

Consider the following choice of 4 half samples:

Halfsample	Stratum 1	2	3	4
1	+	+	+	-
2	+	-	-	-
3	-	-	+	-
4	-	+	-	-

\+ First element (PSU) is selected

\- Second element (PSU) is selected

Balanced Repeated Replication (BRR)

- Square arrays of + and - that define **balanced** (orthogonal) half samples are called **Hadamard matrices**. Placket and Burman (Biometrika, 1946) have tabulated these matrices for H=4,8,12,16,…,200 (all multiples of 4)

- Computer algorithms are available for creating these Hadamard matrices (Stata, WesVar PC), which simplify variance estimation (eliminating covariance terms).

- What if the number of strata is not multiple of 4? In this case only partial balance can be achieved.
  - For H=3 drop one column from the matrix for H=4. For H=5 choose the matrix with H=8 and drop 3 strata (columns).

BRR: Constructing Replicates, Replicate Weights

- Suppose there are H strata with a_h clusters. H=4, a_h=2 in example data set.
- A BRR replicate is constructed by deleting one PSU from each stratum according to the pattern specified in the Hadamard matrix.
  - Example: the first replicate leaves out the 2^{nd} PSU in Strata 1,2,3 and the 1^{st} PSU in Stratum 4.
- The replicate weight for this first replicate multiplies the weights for remaining cases in the half-sample by a factor of 2.

BRR Replicate 1: Data Example

Stratum	PSU (Cluster)	Case	y_i	$w_{i,rep}$
1	1	1	.58	1x2
	1	2	.48	2x2
1	2	1	.	.
	2	2	.	.
2	1	1	.39	1x2
	1	2	.46	2x2
2	2	1	.	.
	2	2	.	.
3	1	1	.39	1x2
	1	2	.47	2x2
3	2	1	.	.
	2	2	.	.
4	1	1	.	.
	1	2	.	.
4	2	1	.47	2x2
	2	2	.50	2x2

BRR: Constructing Replicates, Replicate Weights(2)

- Example: the second replicate leaves out the 2nd PSU in Stratum 1 and the 1st PSU in Strata 2,3,4.
- Again, the replicate weight for this second replicate multiplies the weights for remaining cases in the half-sample by a factor of 2 (nonresponse and calibration adjustments could also be applied).
- Four half-sample replicates are created based on the deletion pattern in the Hadamard matrix

BRR Replicate 2: Data Example

Stratum	PSU (Cluster)	Case	y_i	$w_{i,2,brr}$
1	1	1	.58	1x2
	1	2	.48	2x2
1	2	1	.	.
	2	2	.	.
2	1	1	.	.
	1	2	.	.
2	2	1	.50	2x2
	2	2	.21	1x2
3	1	1	.	.
	1	2	.	.
3	2	1	.44	1x2
	2	2	.43	1x2
4	1	1	.	.
	1	2	.	.
4	2	1	.47	2x2
	2	2	.50	2x2

BRR: Constructing Estimates

$$\widehat{q_r} = y_r^w = \frac{\sum_{i \in rep}^{n_{rep}} y_i \cdot w_{i,rep}}{\sum_{i \in rep}^{n_{rep}} w_{i,rep}}$$

is computed for each of r=1,…,4 replicates

$$\widehat{q_1} = .4708;\ \widehat{q_2} = .4633;\ \widehat{q_3} = .4614;\ \widehat{q_4} = .4692;$$

$$\hat{q} = \frac{\sum_{i=1}^{n} y_i \cdot w_i}{\sum_{i=1}^{n} w_i}$$

the full sample estimate is also computed.

$$\hat{q} = .4737$$

BRR: Estimating the Sampling Variance

$$\mathrm{var}_{BRR}(\hat{q}) = \frac{1}{R}\sum_{r=1}^{R}(\hat{q}_r - \hat{q})^2$$

$$= \frac{1}{4}\sum_{r=1}^{4}(\bar{y}_r - \bar{y})^2$$

$$= .00007248$$

$$se_{BRR}(\bar{y}) = \sqrt{.0000748} = .008515$$

$$CI(\bar{y}) = 0.47375 \pm t_{1-\alpha/2,4} \cdot .008515$$

$$= 0.47375 \pm 2.7764 \cdot (.008515) = (0.45011, 0.49739)$$

Recall that JRR resulted in a 95% CI of (0.4493, 0.4981), resulting in extremely similar inferences.

Rao-Wu (1988) Rescaling Bootstrap: Variance Estimation

- The variance estimate based on b=1,…,B bootstrap estimates (based on with-replacement samples of the same size) is the Monte Carlo approximation:

$$\mathrm{var}_{Boot}(\hat{q}) = \frac{1}{B}\sum_{b=1}^{B}(\hat{q}_{(b)} - \hat{q})^2$$

- For large numbers of bootstrap replicates, i.e. b>>100, a histogram of the bootstrap simulates the sampling distribution of the estimator
  - May be used to examine asymptotic normality assumption
  - May be used to derive asymmetric CIs for population values

Rao-Wu Rescaling Bootstrap: Replicate Formation for Complex Samples

- b=1,…, B bootstrap replicates formed by sampling with replacement (SWR) m_h PSUs from each of the h=1,…H primary stage strata.

- Recommendation (Rust and Rao, 1996) is to set $m_h=a_h-1$ (one less than the number of sample PSUs in stratum h)

- For $m_h=a_h-1$, the bootstrap weight for the selected replicate is:

$$w_{h\alpha i}^{(b)} = w_{h\alpha i} \cdot \frac{a_h}{(a_h - 1)} \cdot r_{h\alpha}^{(b)};$$

$$where: r_{h\alpha}^{(b)} = \text{the count of time PSU } \alpha \text{ is selected in replicate b.}$$

Example Comparison of Standard Error Estimates for Estimates of Population Parameters (source: NCS-R).

Statistic / Method		Estimate	Standard Error
Major depression	TSL	19.1711%	0.4877%
(Prevalence)	JRR	19.1711%	0.4878%
	BRR	19.1711%	0.4896%
Alcohol Dependence	TSL	5.4065%	0.3248%
(Prevalence)	JRR	5.4065%	0.3248%
	BRR	5.4065%	0.3251%
Household Income	TSL	$59,277.06	$1596.34
(Mean)	JRR	$59,277.06	$1596.65
	BRR	$59,277.06	$1595.37

Empirical Comparison of Methods

- Data Set: Health and Retirement Study: HRS Wave 1 (1992)
- Software/Variance estimation methods
  - SRS
  - WesVar BRR and JRR with Replicate Weighting for Nonresponse and Poststratification
  - SAS TSL, BRR, JRR using only full sample weight and program creation of replicates
  - R System: Rao, Wu (1988, 1992) Bootstrap with full sample weight (NR and PS adjustments not redone for each Bootstrap replicate)

Comparison of Variance Estimation Methods: HRS (1992) Means, Full Adult Sample

Variable	n	Mean (Wgt)	SRS (SE)	W-BRR (SE)	W-JKK (SE)	S-TSL (SE)	S-BRR (SE)	S-JKK (SE)	R-BS (SE)
Years of School	9759	12.31	0.031	0.078	0.077	0.077	0.077	0.077	0.076
Body Weight (lbs)	9759	172.35	0.374	0.447	0.432	0.435	0.434	0.435	0.404
Memory Recall (Words in Test 10)	9085	7.57	0.028	0.064	0.064	0.066	0.066	0.066	0.067

Comparison of Variance Estimation Methods: HRS (1992) Linear Regression, Words Recalled.

Coefficients	Coef Est (Wgt)	SRS (SE)	W-BRR (SE)	W-JKK (SE)	S-TSL (SE)	S-BRR (SE)	S-JKK (SE)	R-BS (SE)
INTERCEPT	4.88	0.120	0.170	0.170	0.166	0.170	0.166	0.178
Male	-0.93	0.052	0.063	0.065	0.064	0.064	0.064	0.061
Black	-1.24	0.084	0.089	0.091	0.093	0.094	0.093	0.096
Hispanic	-0.51	0.107	0.140	0.143	0.141	0.141	0.141	0.139
SCHLYRS	0.27	0.009	0.011	0.012	0.011	0.011	0.011	0.012